tf.compat.v1.math.reduce_euclidean_norm Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/reduce_euclidean_norm

> *Computes the Euclidean norm of elements across dimensions of a tensor.*

Function Signatures

```
tf.compat.v1.math.reduce_euclidean_norm( input_tensor,
axis=None, keepdims=False, name=None )
```

Code Examples

```
tf.compat.v1.math.reduce_euclidean_norm(
input_tensor, axis=None, keepdims=False, name=None
)
```


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input_tensor, axis=None, keepdims=False, name=None
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```


```
x = tf.constant([[1, 2, 3], [1, 1, 1]]) # x.dtype is tf.int32
tf.math.reduce_euclidean_norm(x)  # returns 4 as dtype is tf.int32
y = tf.constant([[1, 2, 3], [1, 1, 1]], dtype = tf.float32)
tf.math.reduce_euclidean_norm(y)  # returns 4.1231055 which is
sqrt(17)
tf.math.reduce_euclidean_norm(y, 0)  # [sqrt(2), sqrt(5), sqrt(10)]
tf.math.reduce_euclidean_norm(y, 1)  # [sqrt(14), sqrt(3)]
tf.math.reduce_euclidean_norm(y, 1, keepdims=True)  # [[sqrt(14)],
[sqrt(3)]]
tf.math.reduce_euclidean_norm(y, [0, 1])  # sqrt(17)
```


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[sqrt(3)]]
tf.math.reduce_euclidean_norm(y, [0, 1])  # sqrt(17)
```

Documentation

Computes the Euclidean norm of elements across dimensions of a tensor.

Reduces input_tensor along the dimensions given in axis.
Unless keepdims is true, the rank of the tensor is reduced by 1 for each of the entries in axis, which must be unique. If keepdims is true, the reduced dimensions are retained with length 1.

If axis is None, all dimensions are reduced, and a tensor with a single element is returned.

For example:

Returns